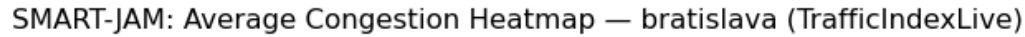




SMART-JAM

- 

- 





Our solution



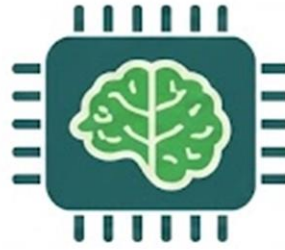
Predict congestion 30-60 minutes ahead



Early warnings for specific areas



Current Time



AI Model



Predicted Congestion

30–60 Minute Forecast



What data we work with



Predict congestion 30-60 minutes ahead



Early warnings for specific areas



Traffic over time



Delay + travel metrics



Jam intensity

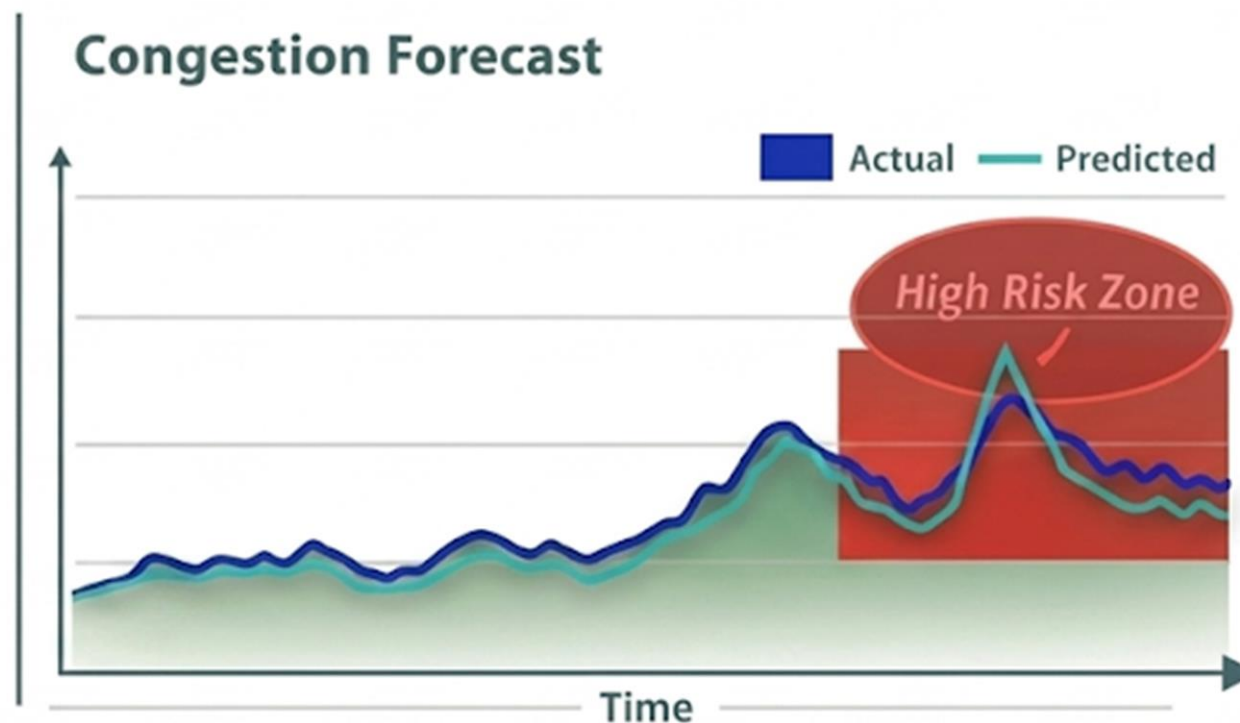
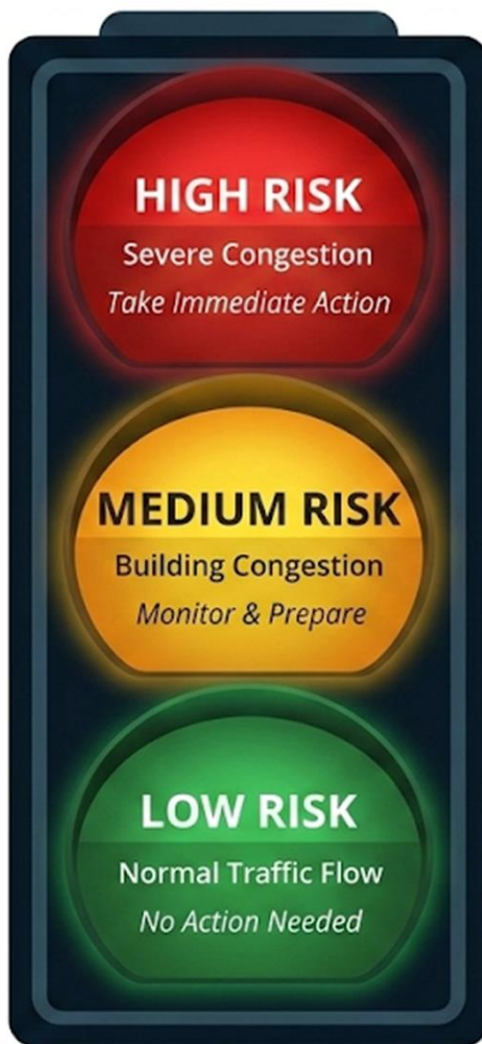


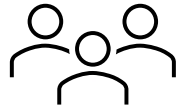
Weekly trends

Time	City	Delay	Index
08:30	Kosice	12.8 min	59%
08:45	Bratislava	17.1 min	62%



Model output





Who benefits ?



Drivers:

- better route choice
- less time in traffic



Cities:

- traffic-light optimization
- routing recommendations



Public transport:

- improved reliability

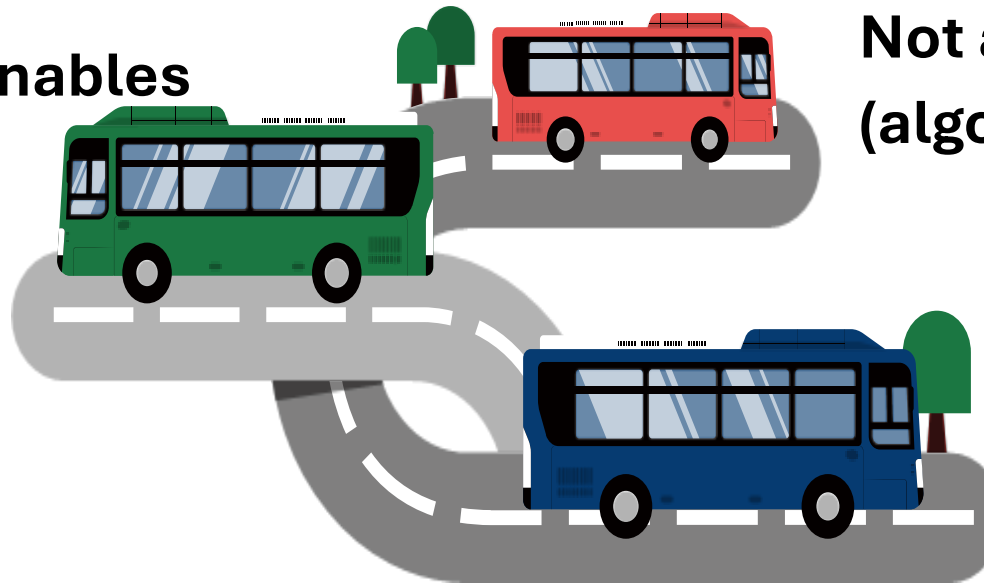
User Requirements



What are user requirements?

What users need the system to do — not how it is built.

**User perspective:
what the system enables**

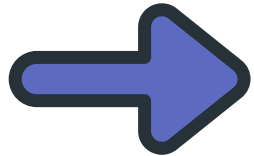


**Not about implementation
(algorithms/architecture)**

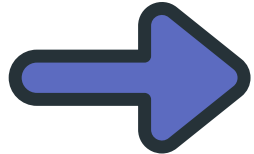
Focus on value & usability



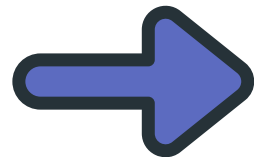
Core requirements?



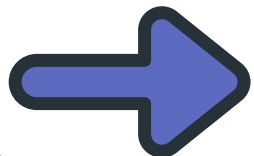
Input



Predict



Risk



Decide





Users

Quick risk + alert → choose an alternative route

Drivers



City Authorities

**Analytics + history + forecast →
manage traffic and policy**



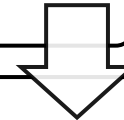
Quality & outcome requirements?

Non-functional needs

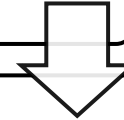
Near real-time, fast response, high availability, scalable, mobile-friendly, and trustworthy.

Why Smart-Jam?

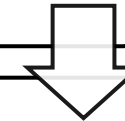
Raw traffic data



actionable insight



less congestion



smarter city movement.

S

Strengths

W

Weaknesses

O

Opportunities

T

Threats



SWOT ANALYSIS

SWOT Overview

Internal

External

S

W

O

T

Strengths

- Practical value for users
- Clear problem focus
- Data-driven approach
- User-oriented design
- Scalable concept

Weaknesses

- Dependence on data quality
- Limited historical data (early stage)
- Model complexity
- Potential performance issues
- User trust challenge

Opportunities

- Growing smart-city initiatives
- Integration with navigation platforms
- Increasing demand for mobility tools
- Expansion to public transport analytics
- Academic and research potential

Threats

- Strong competition
- Data access restrictions
- Rapid changes in traffic patterns
- Privacy and regulation concerns
- Technology expectations

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Threats



- **Strong competition**
- **Data access restrictions**
- **Rapid changes in traffic patterns**
- **Privacy and regulation concerns**
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Model Data Visualisation

Using most up-to date data we plan to determine most efficient routes

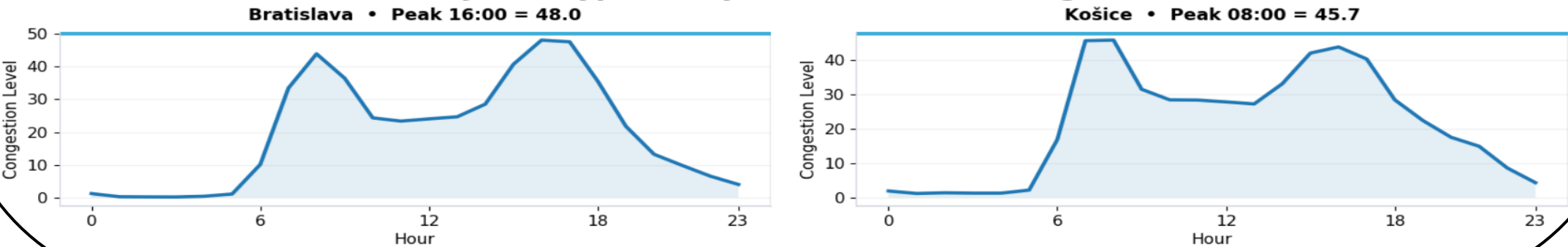
SMART-JAM • Dataset Coverage (Slovakia Focus)

Time range	Countries	SVK cities	SVK rows
2024-10-01 10:01 → 2025-01-13 05:31	55	2	5,185

SMART-JAM • Congestion Level KPIs (Peak vs Off-Peak)

BA peak hour	KE peak hour	BA off-peak	Peak/Off ratio
16:00 • 48.0	08:00 • 45.7	03:00 • 0.2	BA 212.12× • KE 37.02×

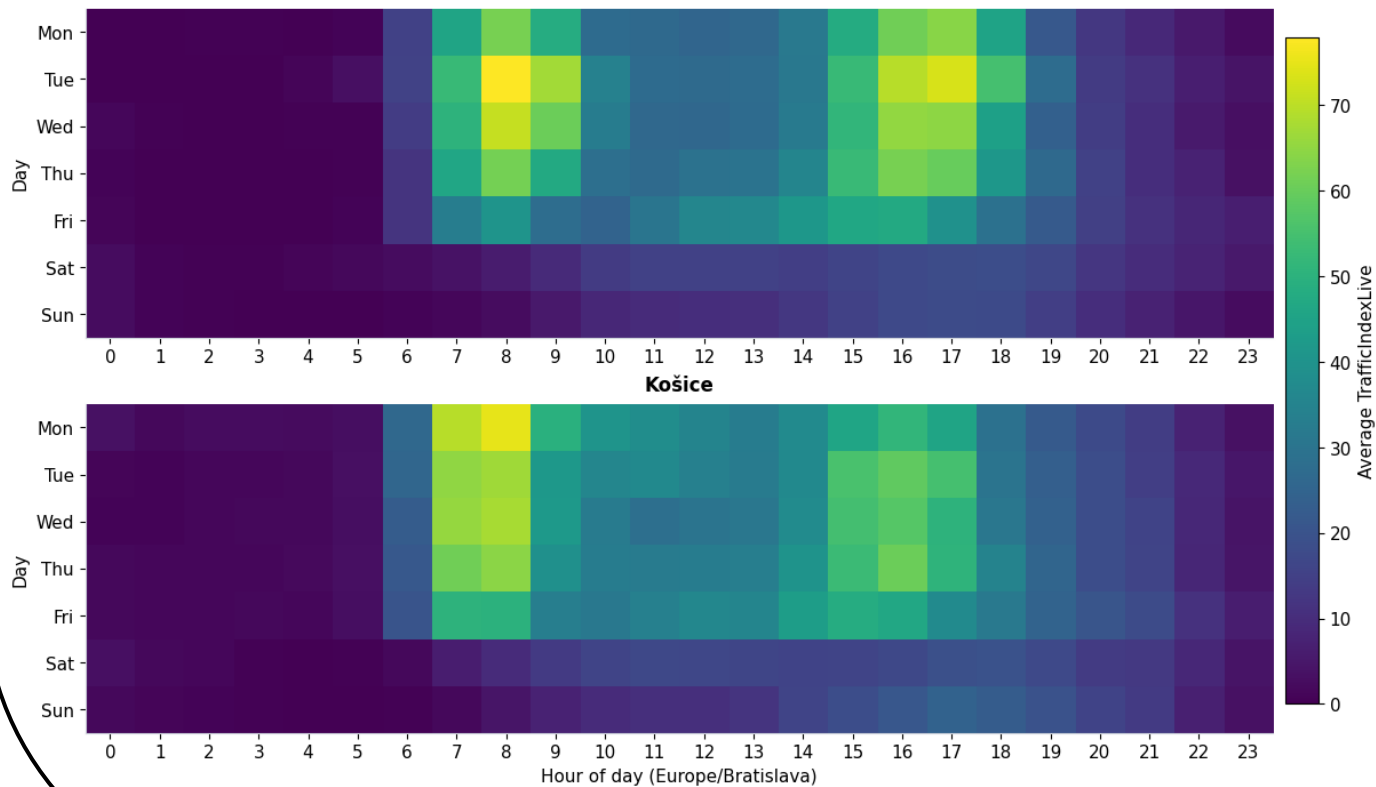
SMART-JAM • Typical Day Pattern (24h) • Congestion Level





Congestion levels

SMART-JAM • Weekly Traffic Heatmaps by City • TrafficIndexLive
Bratislava



(Congestion level of 20 = trips take about 20% longer as opposed to free-flow traffic trips)



Congestion levels

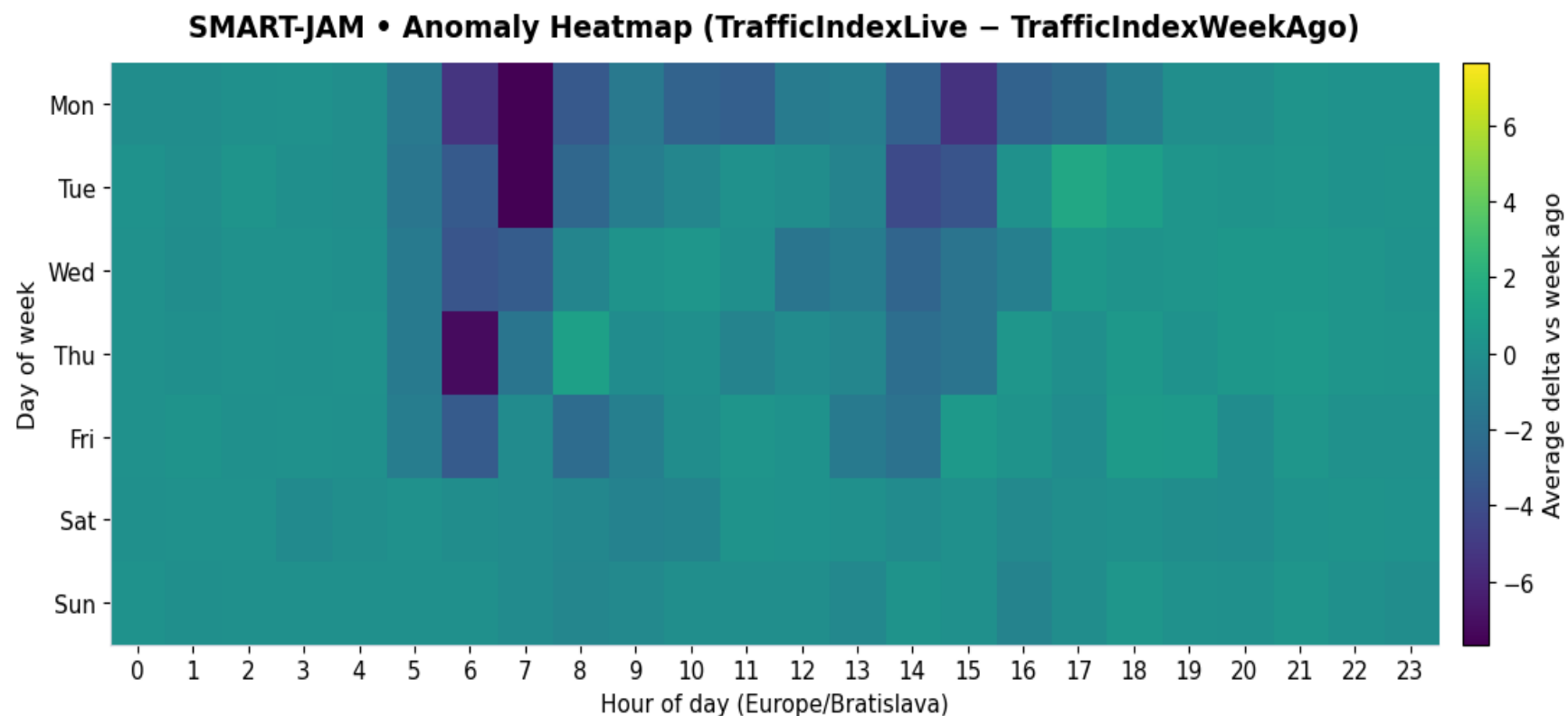
Average over dataset
SMART-JAM • Slovakia Traffic Bubble Map • Congestion Level



Congestion in Košice and Bratislava



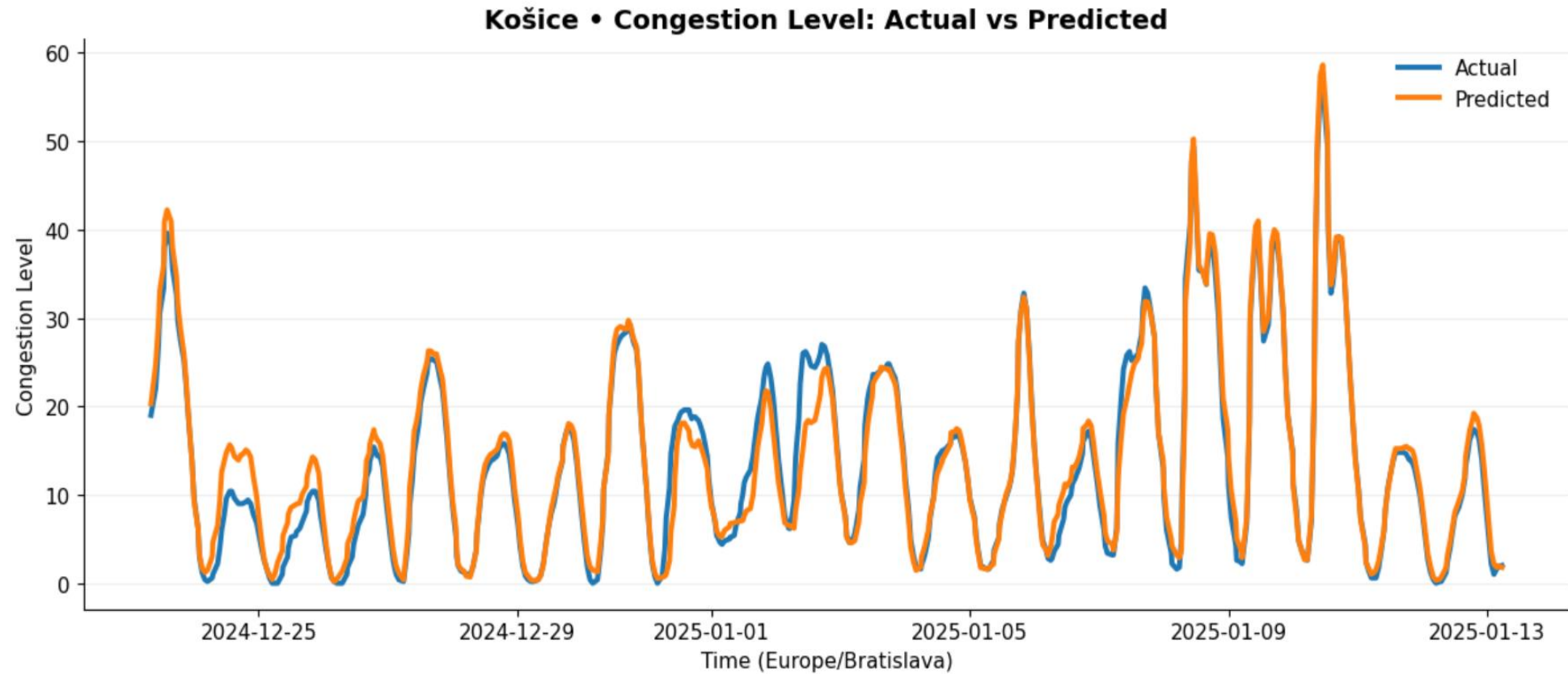
From data to predictions



- Our goal – use real time data to make hasty predictions and satisfy users

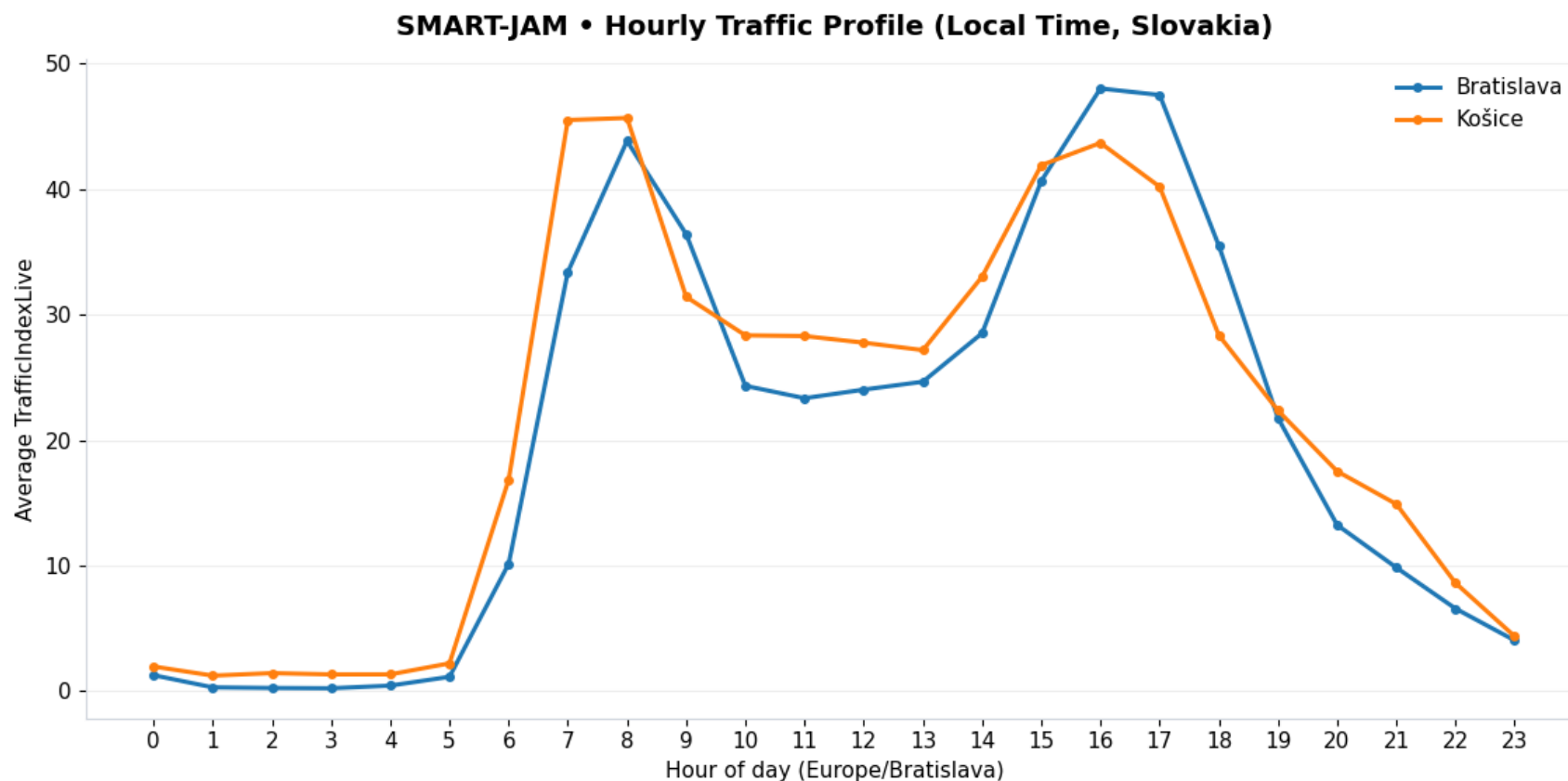


From data to predictions





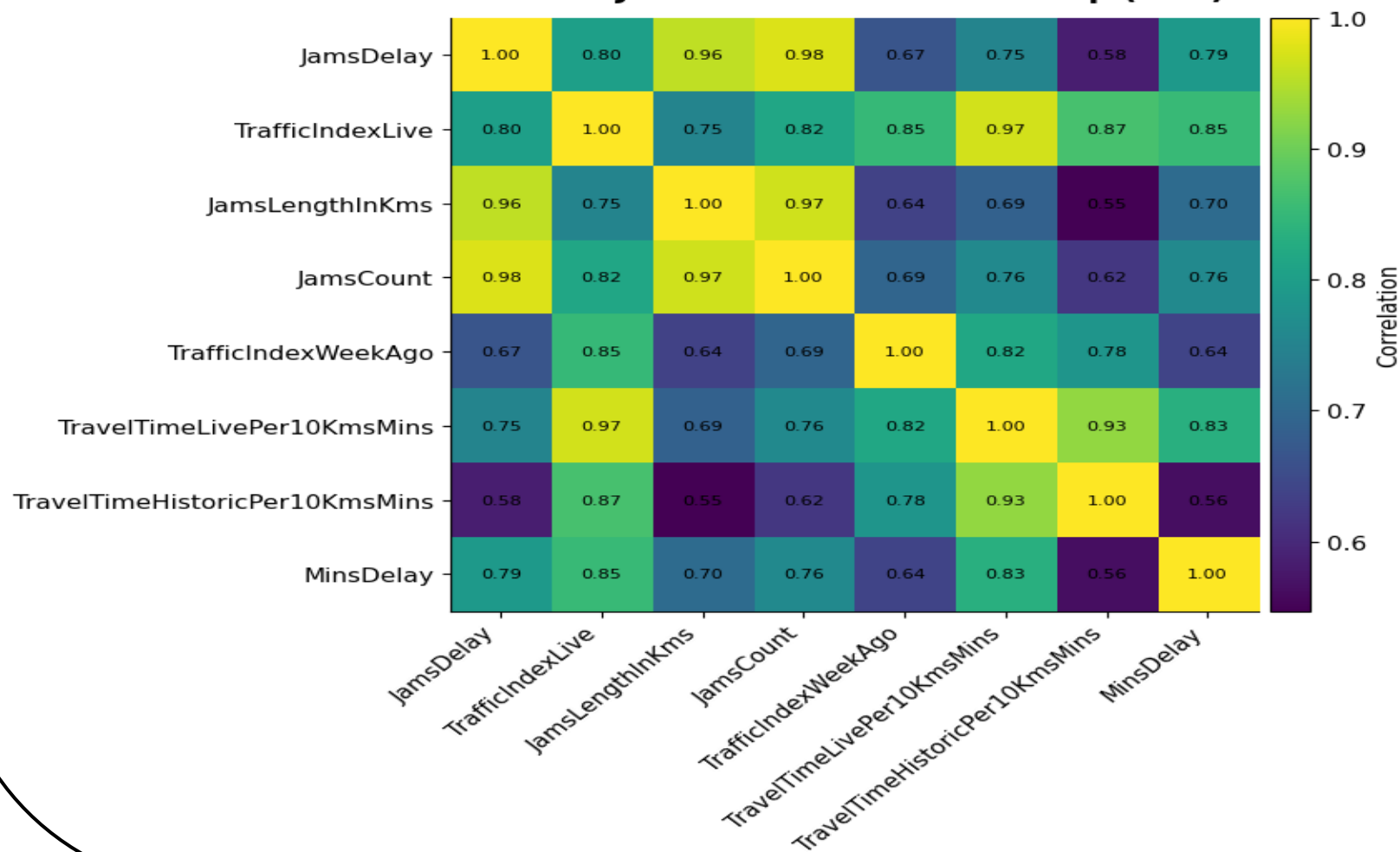
From data to predictions





From data to predictions

SMART-JAM • Correlation Heatmap (SVK)



🔍 Ethical aspects

Privacy Protection

Anonymous driver data

GDPR data processing

No audio recording





Data Security

Encrypted data storage

Authorized access only

Cybersecurity





Transparency

Explainable predictions

Bias prevention

Result reliability

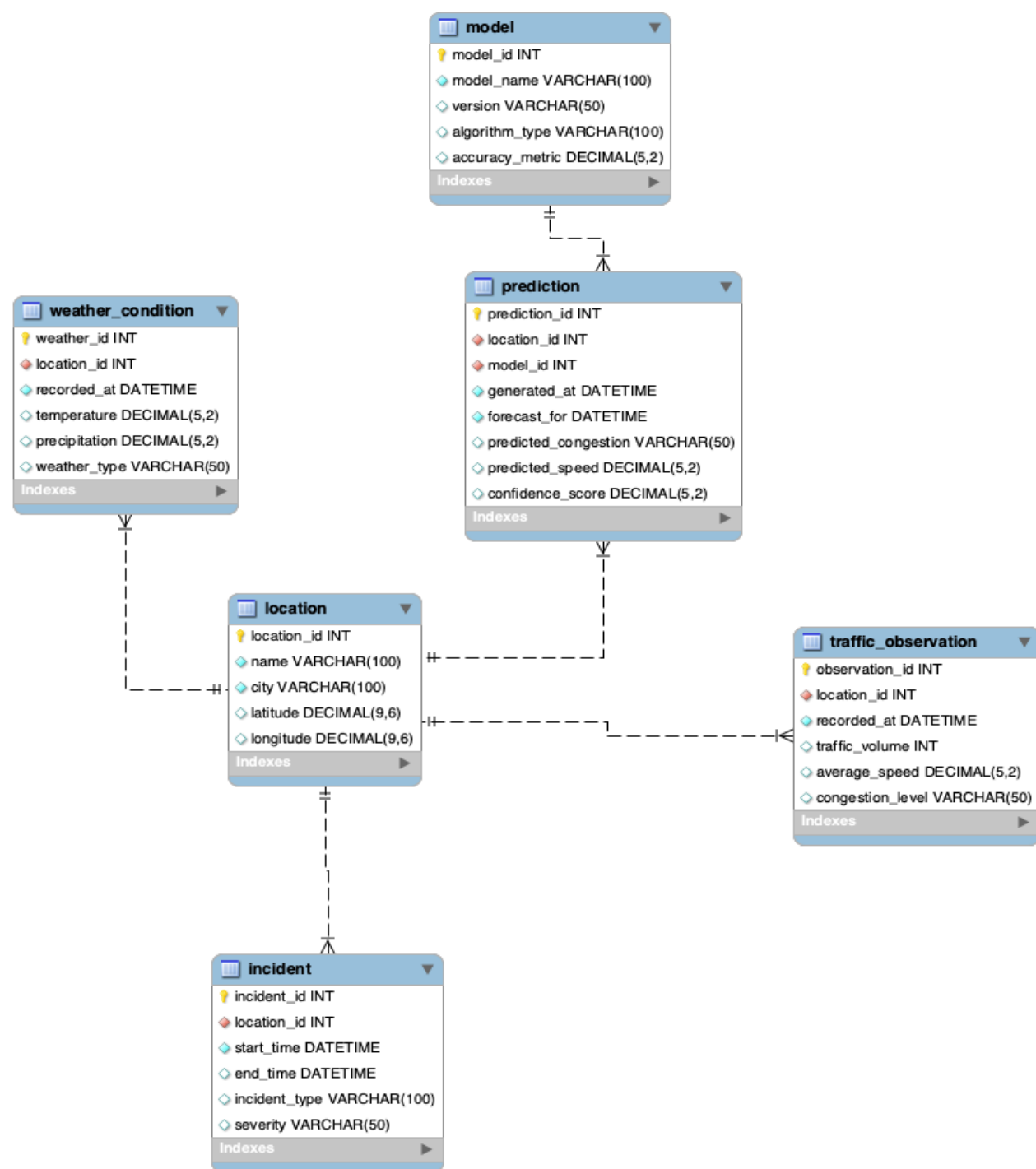


Entity Relationship Diagram



Complete ERD

- The **Smart-Jam** database is designed to support the collection, storage, and analysis of traffic-related data for short-term traffic congestion prediction.
- It stores information about monitored **locations**, real **traffic observations**, **weather conditions**, **incidents** affecting traffic flow, machine learning **models**, and generated **predictions**.



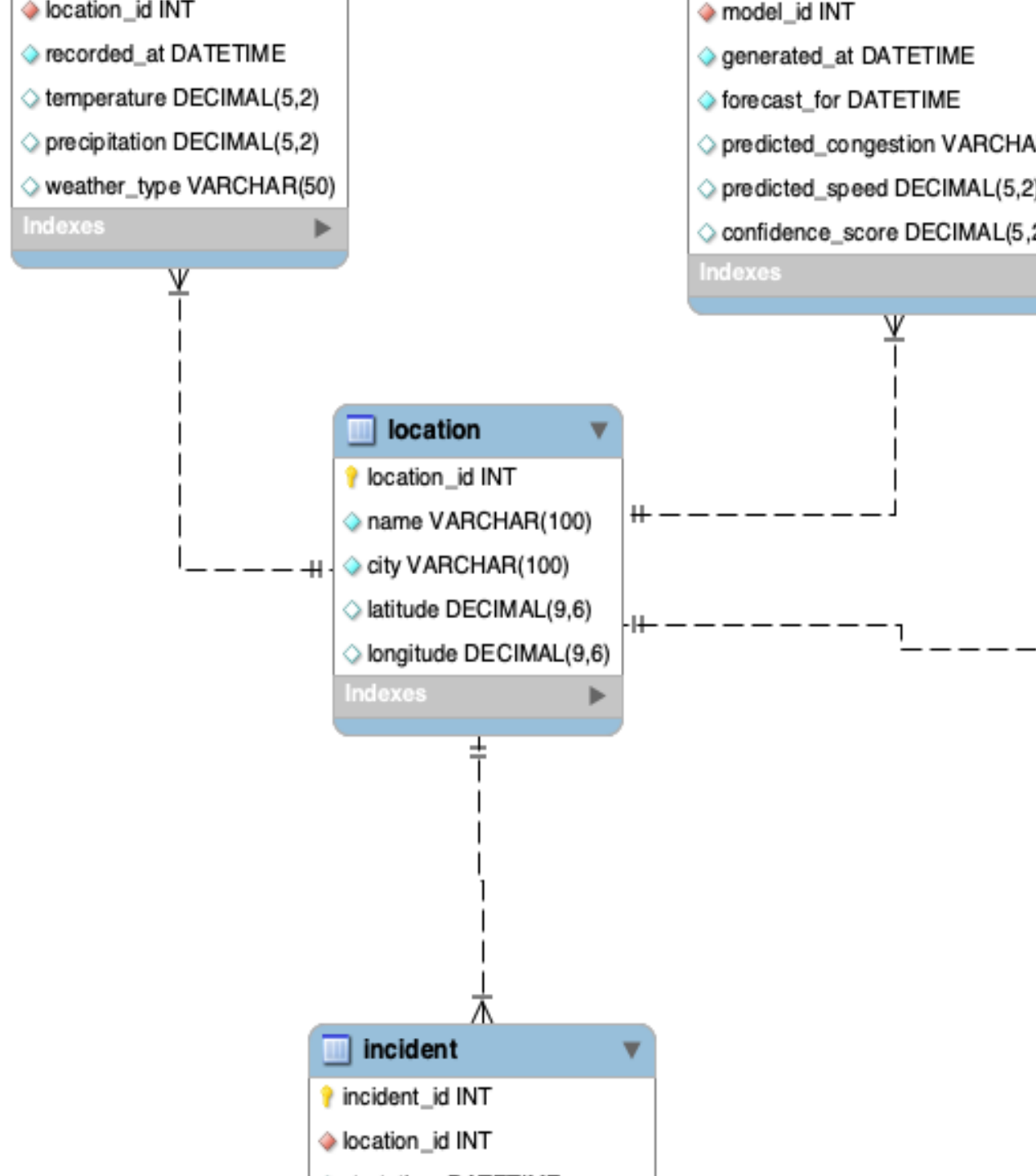


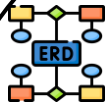
Location

- stores information about monitored road locations, including the location name, city, and geographic coordinates.
- A single location can have multiple traffic observations, weather records, incidents, and predictions.

Cardinality summary >

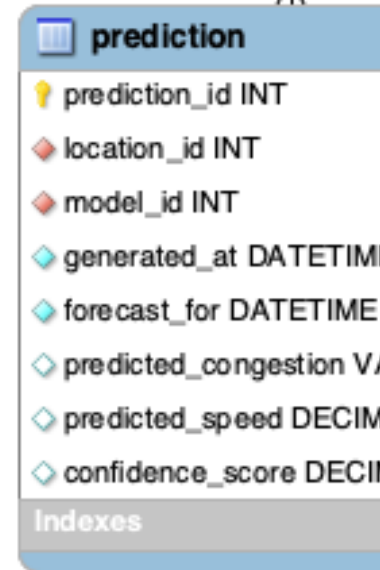
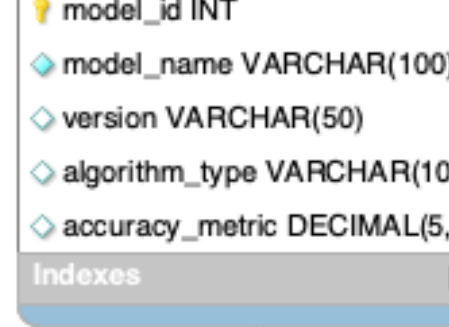
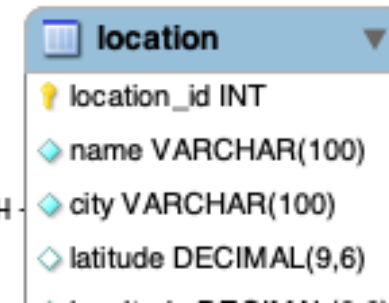
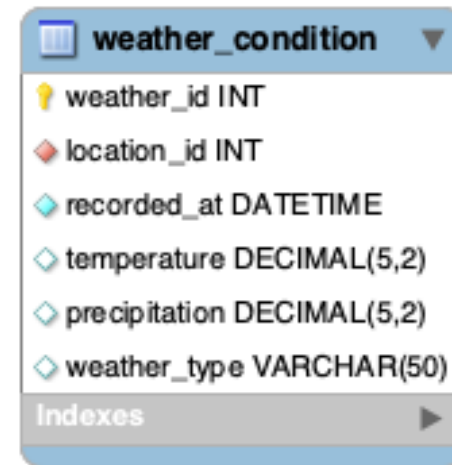
- location → traffic_observation = **1:N**
- location → weather_condition = **1:N**
- location → incident = **1:N**
- location → prediction = **1:N**





Weather condition

- stores weather-related data for a location and time, including temperature, precipitation, and weather type.

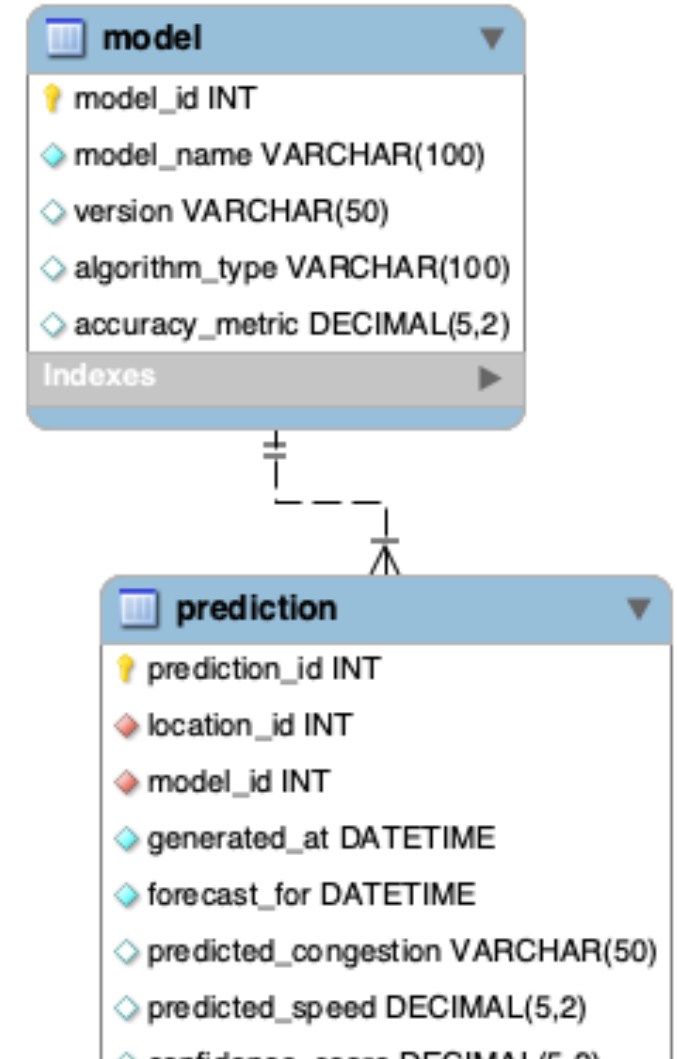
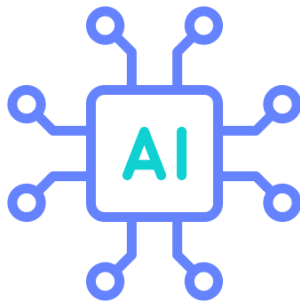




- stores information about the forecasting models used in the system including model name, version, algorithm type, and accuracy metric.

Cardinality summary >

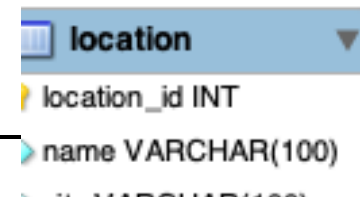
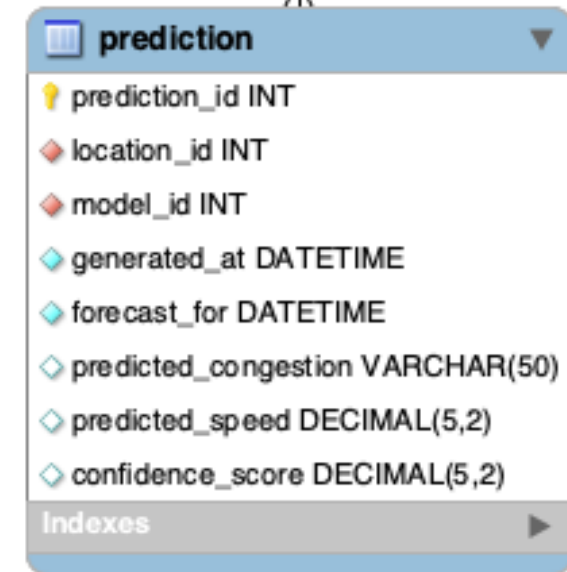
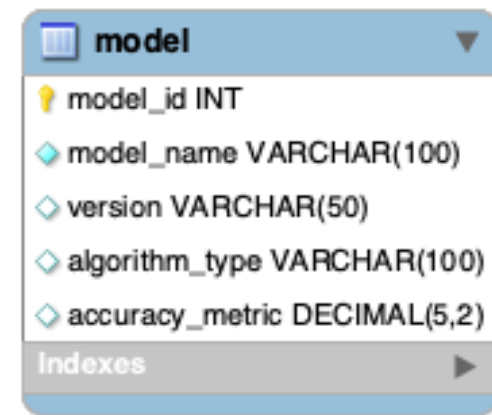
- model → prediction = **1:N**





Prediction

- contains generated traffic forecasts for specific locations and times, including predicted congestion, predicted speed and confidence score.
- each prediction is generated by one specific model.





Traffic observation

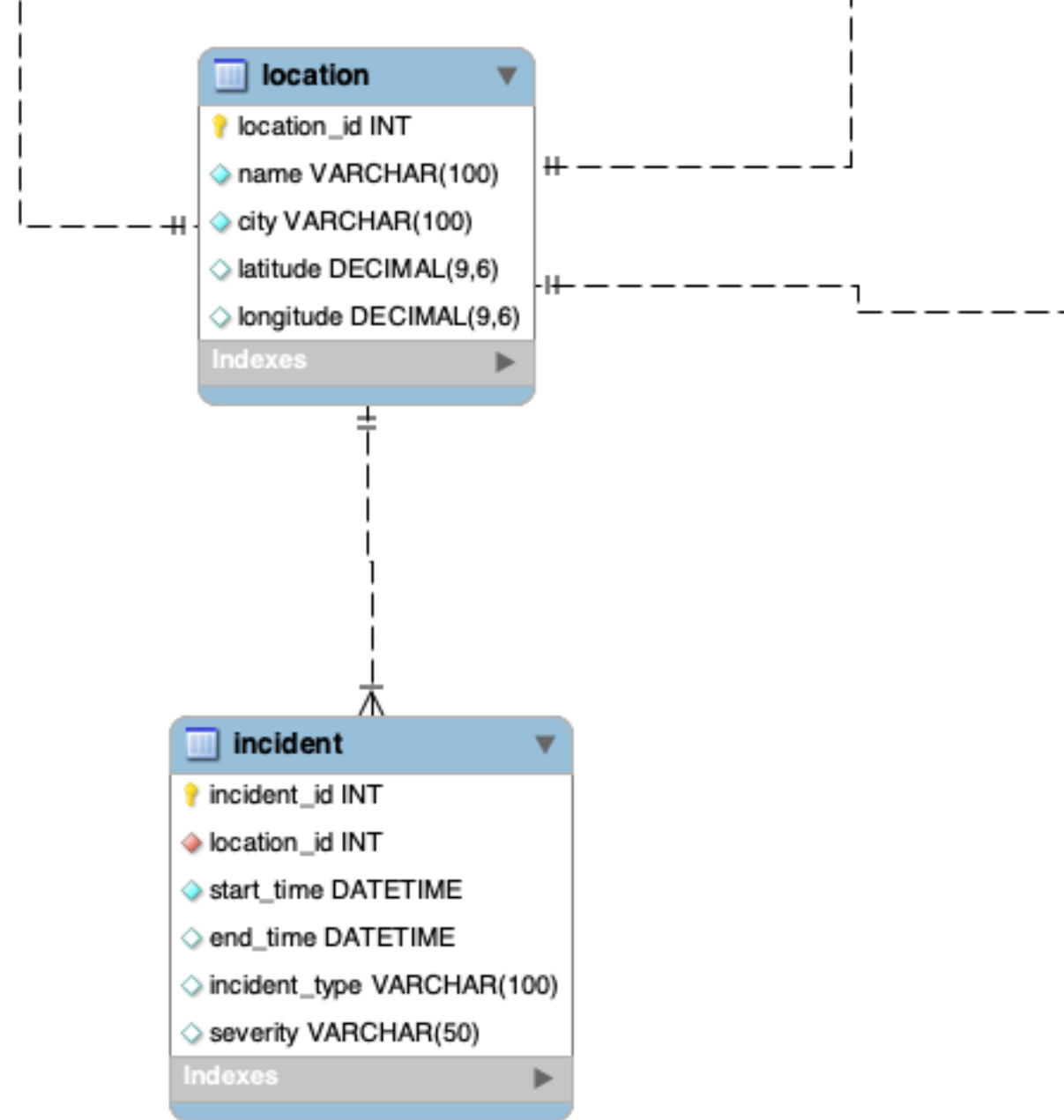
- records real traffic data for a specific location and time, such as traffic volume, average speed, and congestion level.





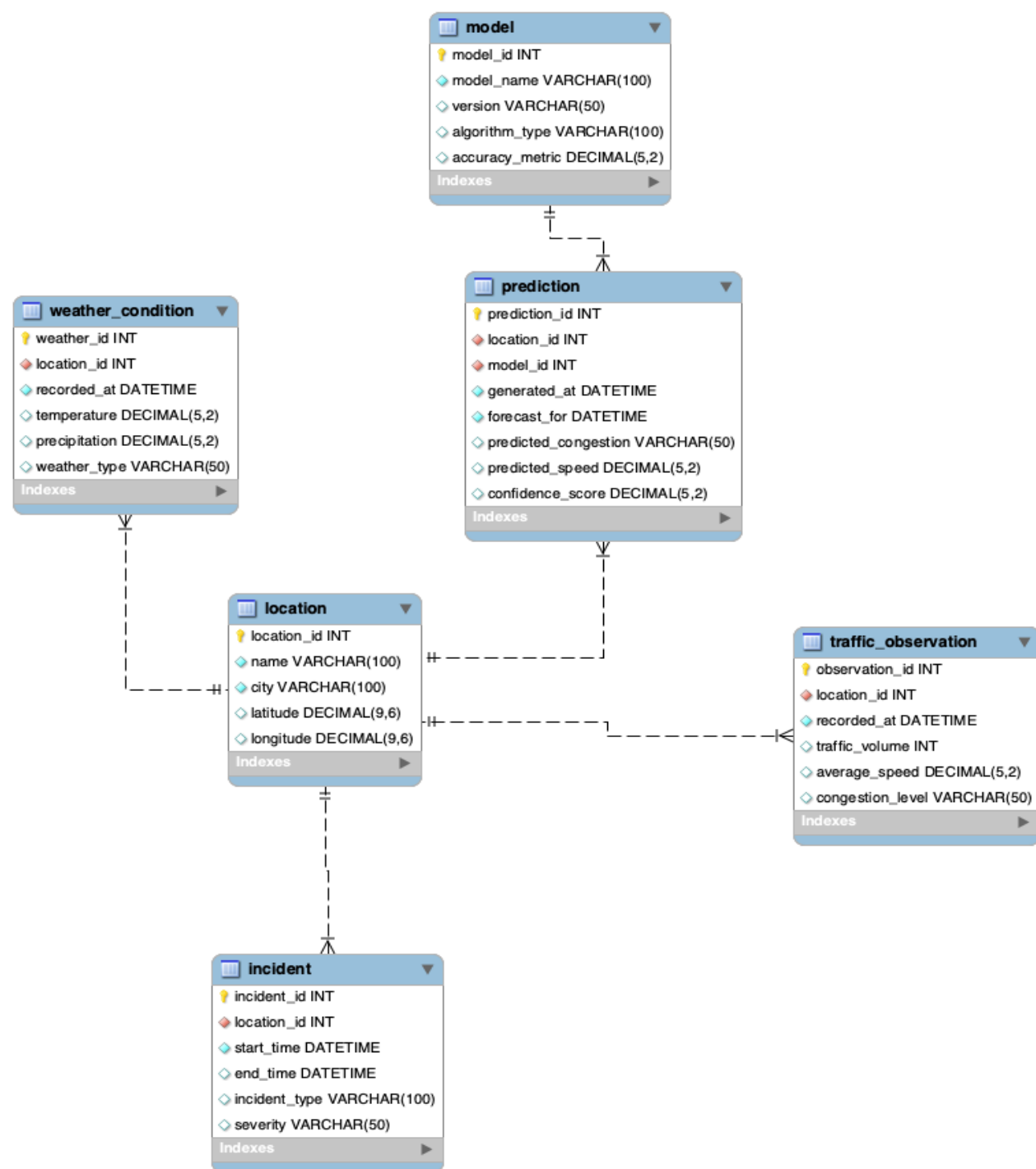
Incident

- contains information about traffic related events that may influence road conditions, such as accidents or roadworks, together with their start and duration.



Database logic

- The relationships between these entities reflect the main logic of the system.
- This structure ensures clear organization of input data and forecasting results while keeping the database simple and suitable for the project scope.
- The database allows the system to analyze current road conditions and produce forecasts for the next 30–60 minutes.



 **Social computing**



WHY SMART-JAM MATTERS ?



Time lost due to traffic in rush hour in 2025

138 hours = 5 days, 18 hours

Similar to 2024.

PREDICT EARLIER. REACT FASTER. PLAN SMARTER.

Traffic congestion is still one of the biggest urban mobility problems. It causes delays, increases emissions, and makes public and private transport less efficient.

Smart-Jam addresses this issue by predicting congestion before it becomes critical.

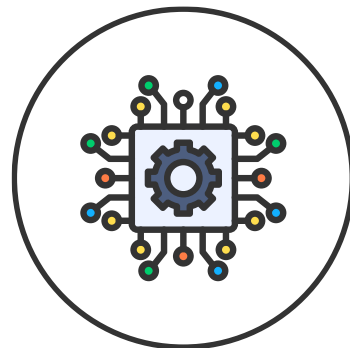
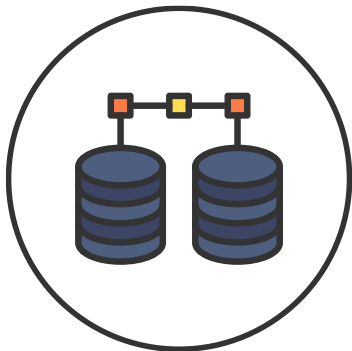
PREDICTING CONGESTION BEFORE IT HAPPENS



MARKET DEMAND

There is clear demand for predictive traffic tools.
Cities and transport operators need solutions that improve planning, reduce delays, and support smarter mobility.
Smart-Jam meets this need through data-driven traffic forecasting.

DATA → PREDICTION → INSIGHT → ACTION





Competition & our advantage

The market already includes strong traffic analytics platforms. Companies such as INRIX, Miovision, and HERE prove that this is a real and growing field. Smart-Jam stands out through focused prediction and easier local deployment.

Big platforms

- BROAD SYSTEMS
- COMPLEX TOOLS
- ENTERPRISE SCALE

SMART-JAM

- FOCUSED SOLUTION
- SIMPLE DASHBOARD
- LOCAL PILOT READY



HOW CAN CUSTOMERS FIND US ?



**AI TRAFFIC PREDICTION
IN KOSICE**



TRAFFIC MANAGEMENT



**URBAN MOBILITY
ANALYTICS**

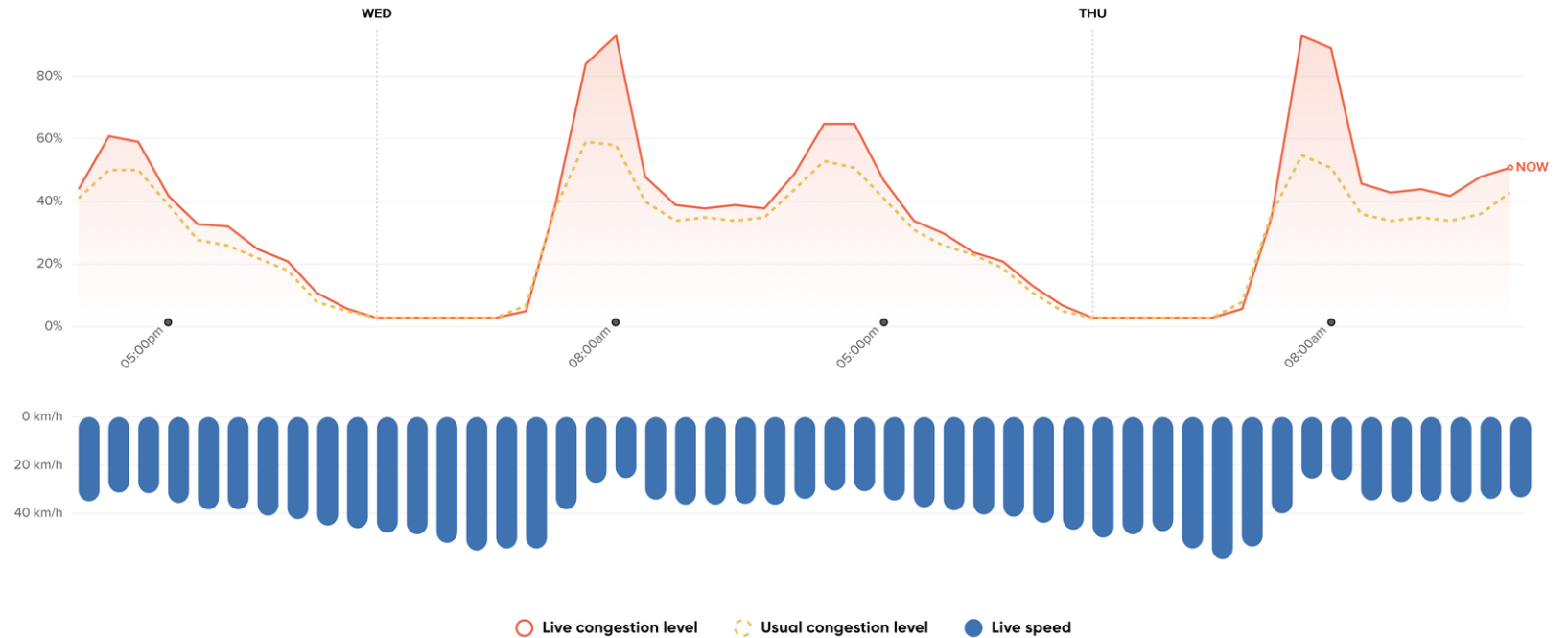
CUSTOMERS USUALLY SEARCH BY PROBLEM, NOT BY BRAND.
THEY LOOK FOR SOLUTIONS SUCH AS AI TRAFFIC PREDICTION,
TRAFFIC CONGESTION FORECASTING, AND SMART CITY
TRAFFIC MANAGEMENT.



PROMOTION & POSITIONING

WE WILL PROMOTE SMART-JAM THROUGH:

- pilot projects
- case studies
- smart-city events
- demo presentations



THANK YOU FOR LISTENING!

[team Smart Jam web](#)

team Smart Jam members :

Faina Petrenko

Mykola Beshynskyi

Richard Buko

Dominik Gmitter

Sources

- [Kaggle – dataset](#)
- [HTTPS://WWW.PINTEREST.COM/PIN/1857741720916171/](https://www.pinterest.com/pin/1857741720916171/)
- [HTTPS://WWW.TOMTOM.COM/SOLUTIONS/GOVERNMENT-AND-PUBLIC-SECTOR/](https://www.tomtom.com/solutions/government-and-public-sector/)
- MA DOKUMENTS
- FlatIcon icons
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